



Every line of IR, and the pass that put it there — the whole `opt` + `llc` pipeline, in one report you can read.

A static, self-contained HTML report over the stock LLVM 22 tools.
No patched LLVM. No server. No build step.

pop rsp;ret.

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github.com/manasghandat/LLVM-Lens

01 Abstract

LLVM-Lens answers a question LLVM makes surprisingly hard to answer: *what did each pass in the pipeline actually do to my code?*

It runs the stock LLVM 22 tools with their own dump flags — `opt` for the middle end and `llc` for the backend — reads the stream they produce, and emits a single static HTML file. Open it in any browser. There is no server, no npm, no build step, and nothing to install on the machine that reads it.

What you get is one card per pass, holding every run that pass made: the before-and-after diff of any one of those runs and nothing else, the control-flow graph on either side of it, its timing, the analyses it ran, and the spill or register-allocation tables it produced if it was a backend pass. Every line of IR and Machine IR in the report carries the pass that wrote it, so you can click any line and walk its whole history back to the input. A flow view shows the entire lane at once, with runs that changed nothing collapsed, so the shape of the pipeline is visible in one screen instead of forty thousand lines of stdout.

You can also load your own pass plugin and have it treated as a first-class citizen of the report, badged and diffed like any stock pass.

LLVM-Lens is our implementation of the brief's LLVM Pass Transformation Analyzer (LPTA).

02 Why we built it

We wanted a good way to visualize the optimization pipeline: every pass that runs, in order, and what each one did to the code.

LLVM will tell you almost everything you need. It just won't tell you in a form you can read.

`opt -print-after-all` on one translation unit of a real C++ file is tens of thousands of lines of IR, in order, with no way to ask it anything. `-print-changed` is better — it only dumps when a pass actually modified the IR — but it tells you *that* a pass changed something, never *what*: to see the change you diff two consecutive dumps yourself, by hand, in an editor. `-debug-pass-manager` gives you the order of the pipeline and the analyses each pass requested, but none of the content. `-time-passes` aggregates a pass's time over the entire lane, so when `SR0APass` ran twenty-eight times you get one number and no way to split it across the runs. And `llc` is a different world again: Machine IR, different flags, different structure, and no correspondence at all between its output and the IR you were just looking at.

So the workflow, for all of us, has been: dump everything to files, pick two, diff them, scroll, lose your place, do it again. It works. It is also why a "why is this loop not vectorized" question takes an hour.

The observation the whole tool rests on

In the second week we noticed something that changed the design. Because `opt` only dumps when a pass has *changed* the IR, the sequence of dumps is not a sequence of snapshots — it is a sequence of *edits*. A pass that produced no dump

produced no change. Which means every line in the final module has exactly one pass that put it there, not probably and not heuristically, but by construction.

That turns "which pass did this?" from a search into a lookup, and it is what makes the blame view well-defined rather than a guess. Everything else in LLVM-Lens serves that one property: if a line knows its author, you can ask the compiler questions instead of grepping at it.

The dumps are already a complete, ordered, line-level record of who changed what. Nobody had joined them up.

03 What we built

A Python package with one command, and a report that is just files.

```
$ llvm-lens examples/json-query/lexer.cpp -o report --passes 'default<02>'
$ open report/index.html
```

There is no custom LLVM pass here and no patched LLVM. LLVM-Lens drives the tools you already have and reads what they already print — a deliberate constraint, because a tool that needs its own LLVM build is a tool nobody runs on a Tuesday afternoon when they actually have a regression to chase.

source	a .c, .cpp, .ll or .bc file
opt	-print-changed=quiet -print-module-scope -debug-pass=manager -time-passes
llc	-print-after-all -debug-pass=Structure -time-passes
parsers	states, timelines, the blame walk, ISEL and asm correlation
report	index.html + manifest.json + one chunk of IR per pass

lane A — middle end, LLVM IR · lane B — backend, Machine IR · the report output is static files

Two lanes. **Lane A** is the `opt` new-pass-manager pipeline over LLVM IR; **Lane B** is the `llc` backend over Machine IR. They stay two tabs but join into one timeline: a synthetic *Input IR* card opens lane A, an *Optimized MIR* card closes lane B, and a dashed edge marks the handoff between them. Read only the middle end, or start at the handoff and read codegen.

It survives a bad run

If `opt` crashes or times out, the report still gets written with whatever was captured up to that point, and says so. A half-answer you can read beats an exception trace, especially when the thing you are investigating is a compiler bug.

04 The report, view by view

Every figure below is a real screenshot of the tool running on a real input, rebuilt immediately before this document was written. The numbers in the prose were read back out of those same report files, not typed from memory.

4.1 The workspace

One screen holds everything: the lane's passes down the left with a churn bar per row, a spine at the very top showing every pass in the lane as a tick sized by how much it changed, the selected pass's own functions below that, the panes in the middle, and a log and analysis drawer along the bottom. The coloured marks beside each pass name are that pass's line churn — green added, red removed — so the rail doubles as a density map of where the work happened.

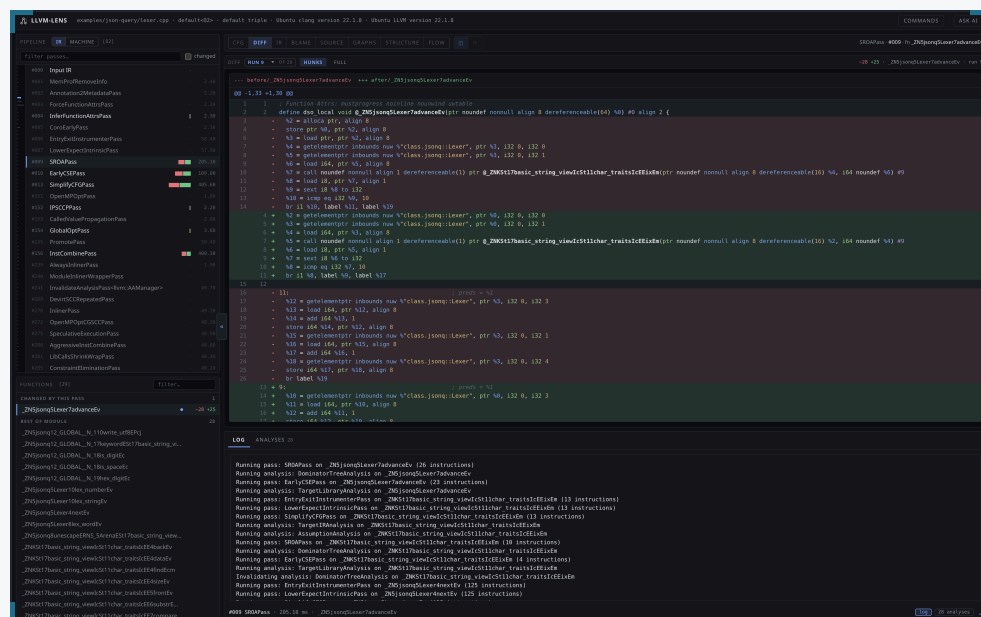


FIGURE 1 – THE WORKSPACE

SROAPass, card #009, run 9 of 28, on `Lexer::advance`. 168 cards in this report — 82 middle-end, 86 backend — covering 350 pass runs, and 22 of the middle-end cards changed something. A card is one pass however many times it ran: SROAPass is a single row in the rail for all 28 of its runs. The menu beside DIFF says which run the panes are reading, and it opens on the last one, because that is the state the card leaves behind. Card numbers are the lane position of the pass's *first* run. The drawer below says this run touched one function, `advance`.

4.2 Diff — one run, and nothing else

This is the view that replaces hand-diffing two dump files. It is the before-and-after of a single pass run, unified, with the churn counted in the header. The diff is exactly one run's work — never a pass's cumulative effect, and never the four other things that happened between two arbitrary dump points. The picker in front of it is what keeps that true once a pass has run many times: it names the run on screen, and the header repeats it.

Choosing an earlier run shows *that* run's diff, against the state it actually replaced. That is not the same as the state before the pass's previous run, because

everything else in the lane ran in between — which is why a run that skipped a function has no diff to show for it and says so, pointing at the runs that did move it.

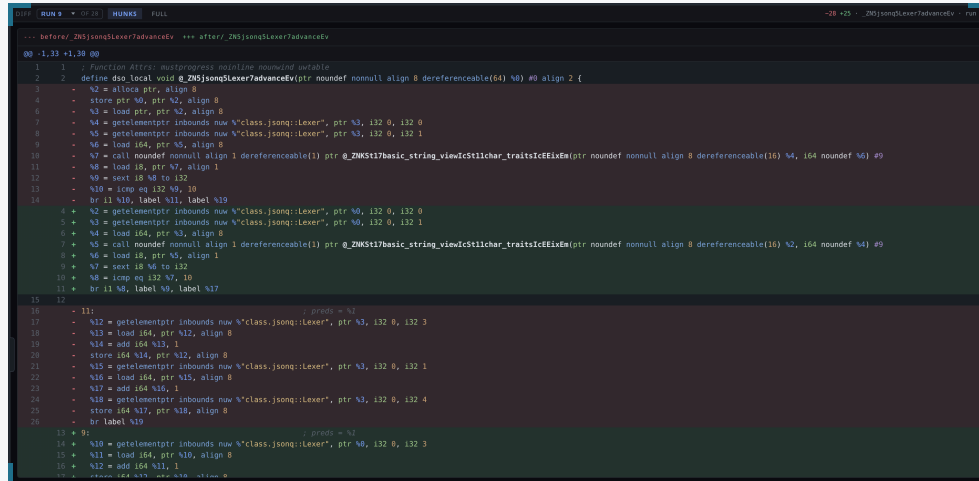


FIGURE 2 – THE DIFF, +25 -28

SROA doing what SROA does: `%2 = alloca ptr`, the store of this into it, and the load back out are all gone, and every later use now reads `%0` — the argument register — directly. The `getelementptr`s that follow change from `ptr %3` to `ptr %0`. Nothing else in the function moved. HUNKS and FULL switch between the changed regions and the entire function, so you can check that claim rather than trust it.

4.3 Blame — who wrote this line

Every line of IR knows the pass that last wrote it. Blame shows that as a gutter down the left: the ordinal of the run that produced the line, or `in` if it has survived untouched since the input, in eight rotating tints so a run's edits read as one block of colour rather than a scatter.

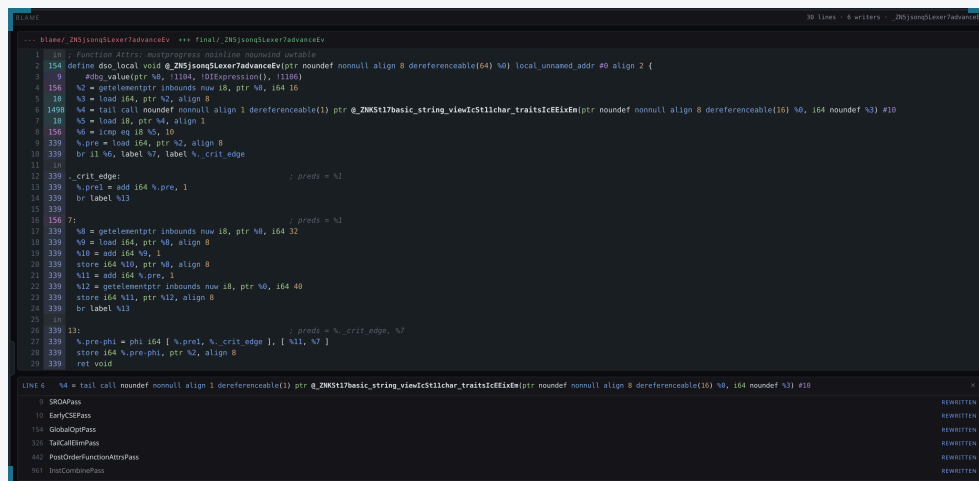


FIGURE 3 – THE GUTTER, AND ONE LINE'S WHOLE HISTORY

`Lexer::advance` after the whole `default<02>` pipeline: 30 lines, 6 writers. The `in` rows are the lines no pass ever rewrote — carried through from the input. Everything else names its author, a run ordinal into the lane. Click any row and the lineage opens below it: every pass that touched that line since the input, oldest first, each tagged **created**, **rewritten** or **renamed** — a distinction that keeps the chain readable, since SSA renumbering turns `%3` into `%7` constantly and that is not a rewrite. Each entry jumps to that pass's card.

The chain in that figure belongs to line 6, the `basic_string_view::operator[]` call, and all seven of its entries are rewrites:

RUN	PASS	WHAT IT DID TO THAT LINE
9	SROAPass	rewritten
10	EarlyCSEPass	rewritten
154	GlobalOptPass	rewritten
326	TailCallElimPass	rewritten
442	PostOrderFunctionAttrsPass	rewritten
961	InstCombinePass	rewritten
1498	PostOrderFunctionAttrsPass	rewritten

Look at what is *not* in that list: a **created**. Line 6 came out of `clang -O0` and no pass has ever introduced it — seven have rewritten it in place, and the last of them, the attributes pass at run 1498, is the one the gutter names. The opposite case sits four rows further down, the `br i1 %6, label %7, label %._crit_edge` at line 10: created by `GVNPass` run 339 and untouched ever since, a chain one row long. Both shapes are answers, and telling them apart by hand means diffing forty thousand lines. Here it is a click.

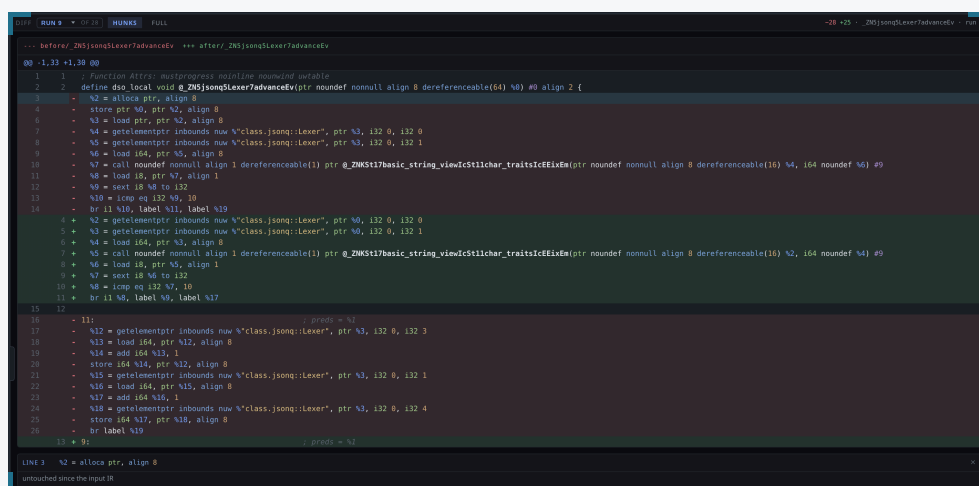


FIGURE 4 – THE QUESTION, ASKED WHERE IT COMES UP

Blame is reachable from the Diff view too, which is where you actually want it: this is the line SROA deleted, `%2 = alloca ptr, align 8`, and the answer is *untouched since the input IR* — it came from `clang -O0` and SROA is the first thing to touch it. Clicking a *green* line instead walks forward from wherever that line was created.

4.4 Flow — the shape of the pipeline

The diff view answers questions about one pass. The flow view answers the question you have before you know which pass to look at: *where in this pipeline is anything happening?* It lays the lane out as a serpentine, one node per pass, and collapses consecutive passes that changed nothing into a single `N` passes · unchanged span. Those collapsed spans are most of the pipeline, and hiding them is what makes the whole thing fit on a screen. Because a node is a pass rather than a run, one node carries everything that pass did in the lane — `SROAPass` is one box holding all 28 of its runs, at its first run's ordinal.

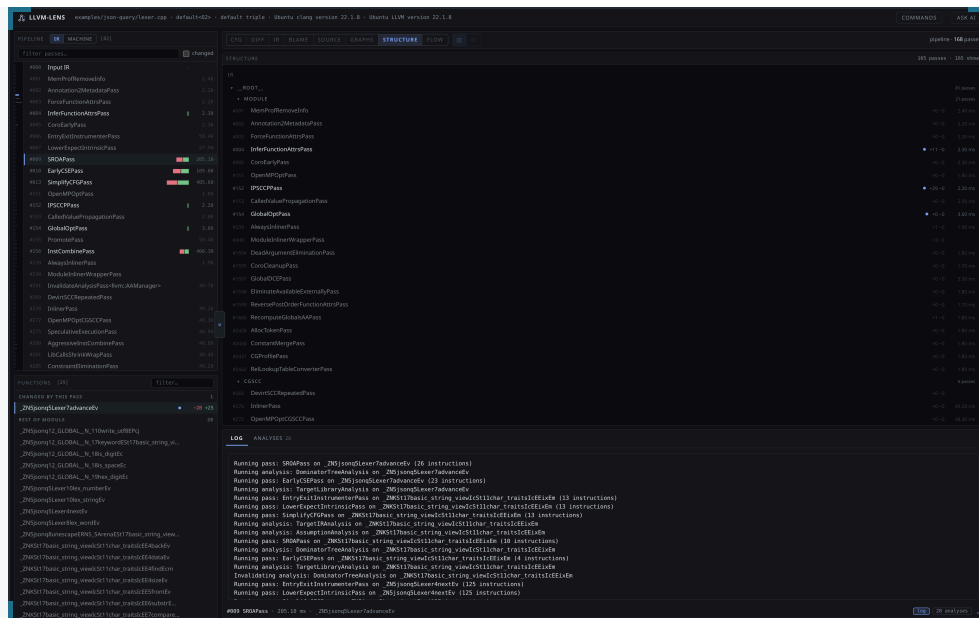


FIGURE 6 – THE HIERARCHY, 165 PASSES SHOWN

The tree is the pass manager's own nesting rather than a flat list – Module, CGSCC, Function and Loop for the middle end, the machine lane's module and function managers beneath, Function 45 passes and Loop 9 passes visible as their own groups. Every leaf carries the analyses it ran and the cached results it invalidated, then its time, and clicking a leaf opens that pass. Read the two columns together and the cost model shows up: #004 InferFunctionAttrsPass reads +11 -0 at 2.30 ms – eleven analyses run, nothing thrown away – while #002 Annotation2MetadataPass reads +0 -0 at 2.20 ms, so the second it costs buys nothing and forces no recomputation downstream.

4.6 Control-flow graphs, before and after

Each card carries a CFG of the selected function on both sides of the pass. Entry block outlined green, back edges red so loops are visible at a glance, block detail on selection. The zoom readout is live – the pane fits the graph, and you zoom from there.

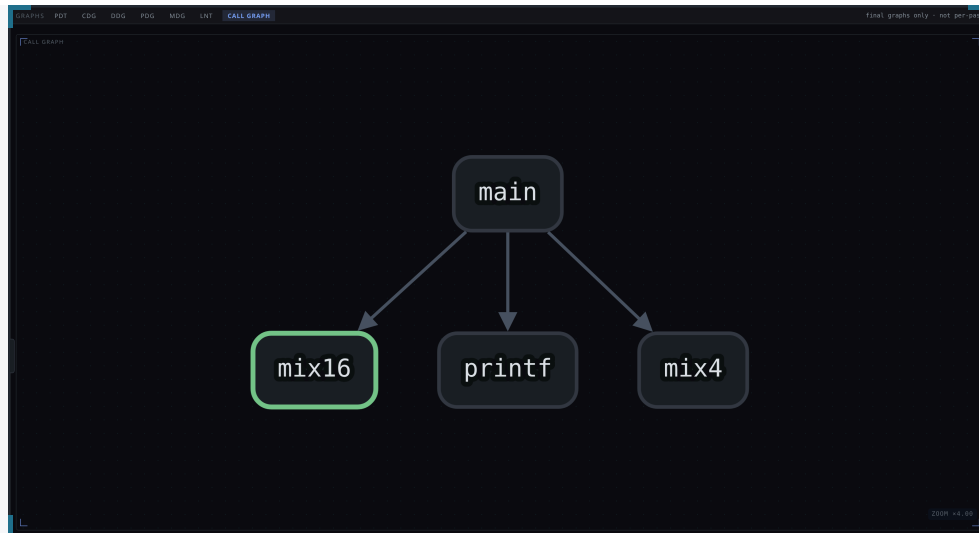


FIGURE 8 – THE CALL GRAPH, AND THE SEVEN GRAPHS ON OFFER

The register-pressure sample, whose four functions fit at **ZOOM ×4.00**: one node per function, one edge per call site, so `main` calls `mix16`, `mix4` and `printf`. The chip row is the honest list – seven graphs, all computed from the final module. This is also the pane where the honest limit shows: these are *final-state* graphs, not per-pass ones, and a per-instruction graph on a real function is unreadably dense – `Lexer::next` fits to ×0.18 and reads as a smear. The report does not pretend otherwise, which is why pass-level questions are answered by Diff, Blame and Flow instead.

4.8 Source – IR line back to C++ line

When the input carries debug info, every IR line maps back to the C++ line it came from, and the header says how many of them mapped. With inlining in play this is the only practical way to see which source a piece of IR actually belongs to.

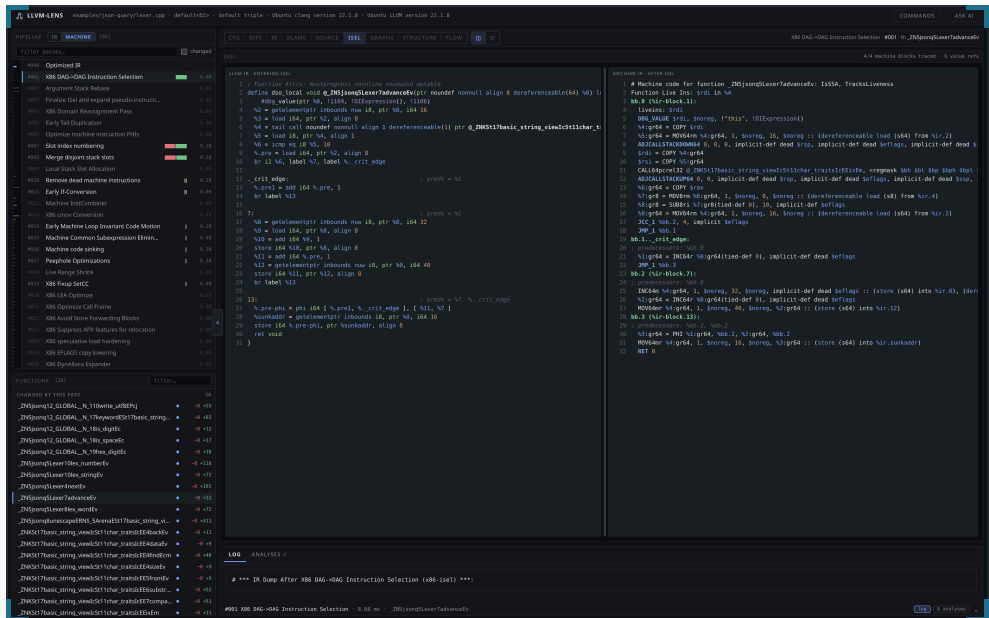


FIGURE 10 – INSTRUCTION SELECTION, 4/4 MACHINE BLOCKS TRACED • 6 VALUE REFS

Left, the IR entering selection; right, the Machine IR leaving it. The pairing is not a guess about ordering – llc names each machine block after the IR block it came from (bb.2 (ir-block-7)), and value operands carry %ir.N references, so %7:gr8 = MOV8rm ... (dereferenceable load (s8)) from %ir.4) really does point at a specific IR line. The header counts how much of the mapping was recovered, so you can tell a complete answer from a partial one.

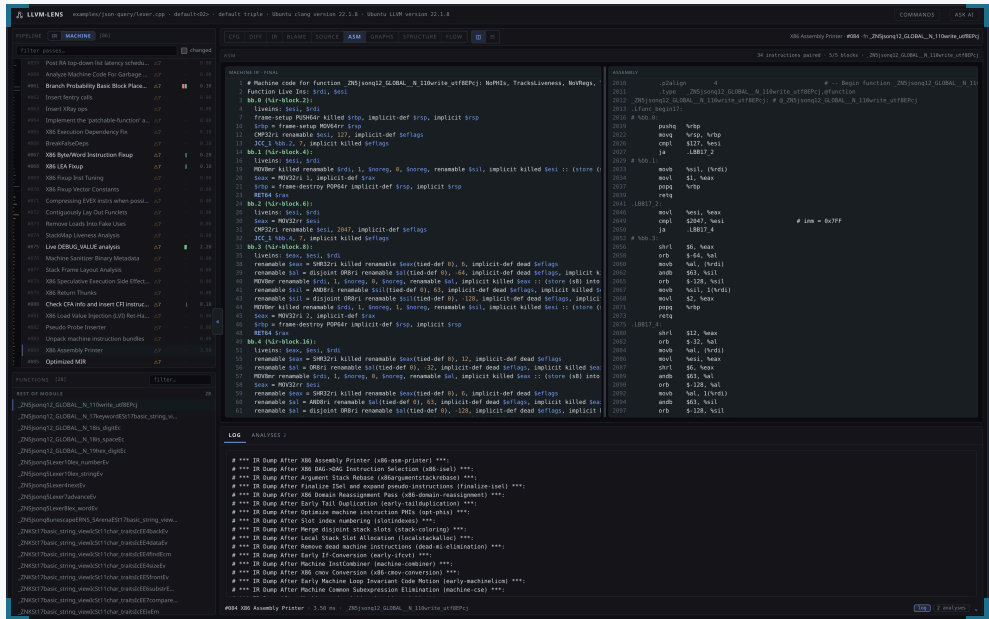


FIGURE 11 – FINAL MACHINE IR AGAINST THE ASSEMBLY PRINTED FROM IT 34 instructions paired • 5/5 blocks. The alignment is a longest common subsequence over an opcode-compatibility predicate – JCC_1 against ja, MOV64mr against movq – so it survives the assembler spelling things differently, and it reports how much it could not pair instead of silently dropping it.

4.10 Spills and registers

When the allocator runs out of registers it spills, and spilled code is usually the answer to "why is this slow". The backend passes carry their own drawers: a spill

table naming every slot and the block it spilled in, and a register map from virtual register to physical register or stack slot, per function.

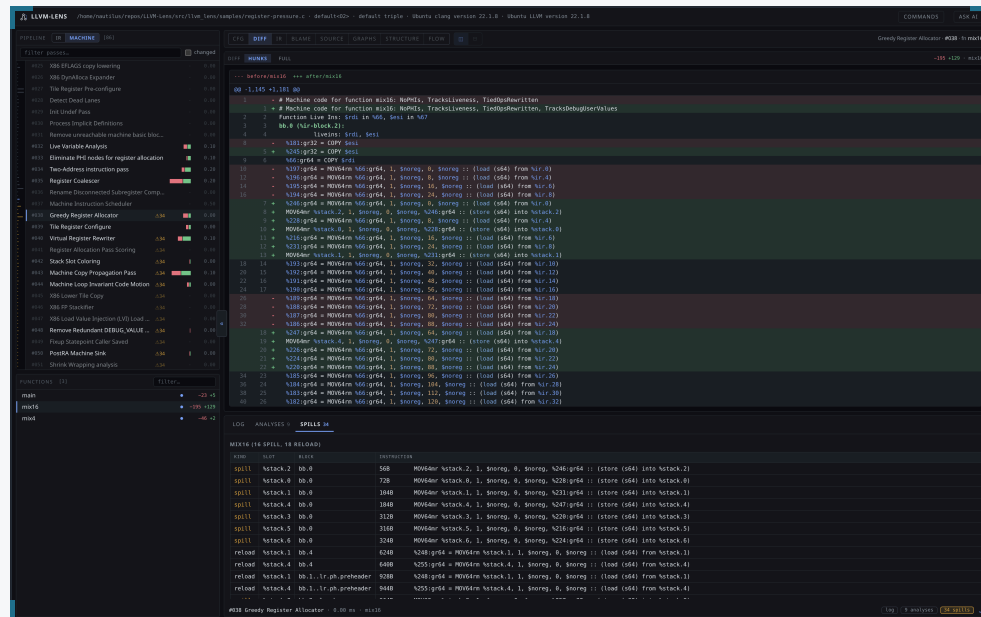


FIGURE 12 – MIX16 , 16 SPILLS AND 18 RELOADS

Greedy Register Allocator – the card is badged with the count (34) and so is every card downstream of it, because the count is a property of the function after allocation, not of one pass. The diff above shows `$gr64` virtual registers turning into `MOV64mr %stack.N, ...` stores; the **Spills** drawer below names the slot, the block (`bb.0` , `bb.4` , `bb.1..lr.ph.preheader`) and the exact instruction, which is the difference between knowing there are 34 spills and knowing where to look.

The companion drawer is the register map on **Virtual Register Rewriter** , the pass where virtual registers become real ones: for `main` it is nineteen rows, `%1` → `$rax` , `%5` → `$rax` , `%3` → `$noreg` . Switch the function to `mix4` in the same report and it is seven rows, every one a physical register, no stack slot anywhere. Same report, same pipeline, two outcomes – which is the whole reason that sample exists. Switch to the **Spills** drawer on `mix16` and the same contrast is there in the numbers: 34 spills against none.

4.11 Your own pass, as a first-class card

A tool for reading LLVM's pipeline is only half useful if it stops at the edge of the passes LLVM ships. Pass `--load-pass-plugin` and `--custom-pass` and your pass appears in the rail, in the flow, in the structure tree and in the blame chains, badged so it is never mistaken for a stock one. Because `opt` dumps custom passes less predictably than built-in ones, the tool forces a dump for them.

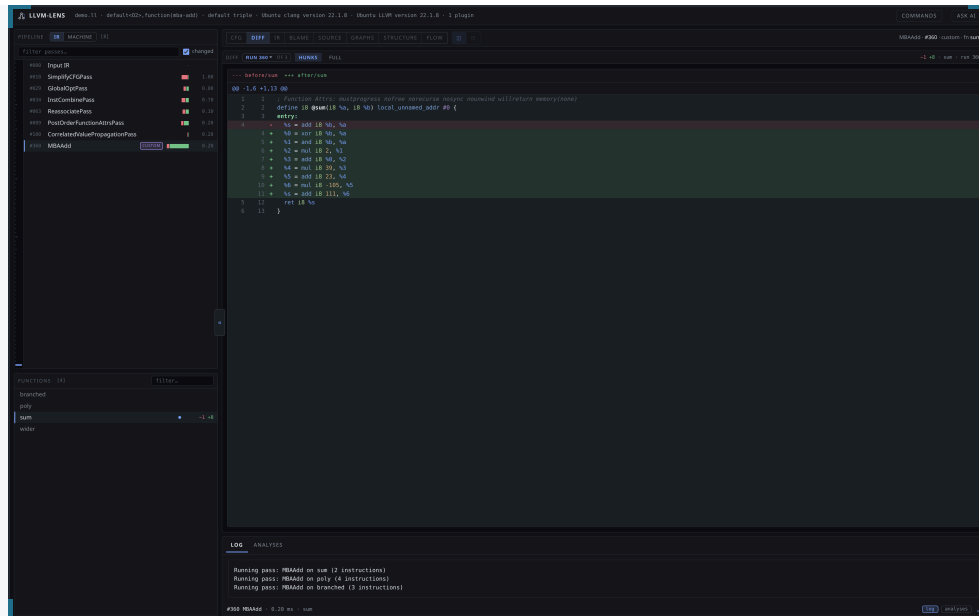


FIGURE 13 – A CUSTOM PASS, +8 -1, ON SUM

MBAAdd rewrites `a + b` into a mixed boolean-arithmetic equivalent: `%s = add i8 %b, %a` becomes eight instructions ending `%s = add i8 111, %6`. Same value, unrecognisable shape – the kind of transform the blame chain is there to survive. The header reads `1 plugin`, the rail badges the card **custom** in purple, and the context line reads `MBAAdd · #360 · custom`. Note the run number: each of its three runs rewrote a *different* function, and the card opens on its last, so this figure was taken on run 360 with the run picker. Only one of the four functions in the module changed in any one run; the pass is selective, and the card shows that rather than implying otherwise.

The same pass appears in the structure tree as `#360 MBAAdd`, badged `custom`, sitting inside `Function 5 passes` where a function pass belongs rather than alongside the module passes. And it is one purple node in the flow view – one node for the pass, however many runs it made, which is the same rule every other pass in that view follows. The plugin was wired entirely by a committed `llvm-lens.yml` beside the example, so running `llvm-lens demo.ll` in that directory needs no flags at all; the config file that did it is named in the report's own metadata, and it is worth a look on its own – [section 05](#) takes it apart.

4.12 Scale

One translation unit is the common case, but the tool is not limited to it. Link seven translation units of a real program into one module and the report grows to match – same views, same blame, 99 functions instead of 29, and the same 168 cards, because the pipeline is the same pipeline. What grows is inside them: the `SR0APass` card that holds 28 runs over one file holds 98 over the linked module, 830 runs in the lane against 264.

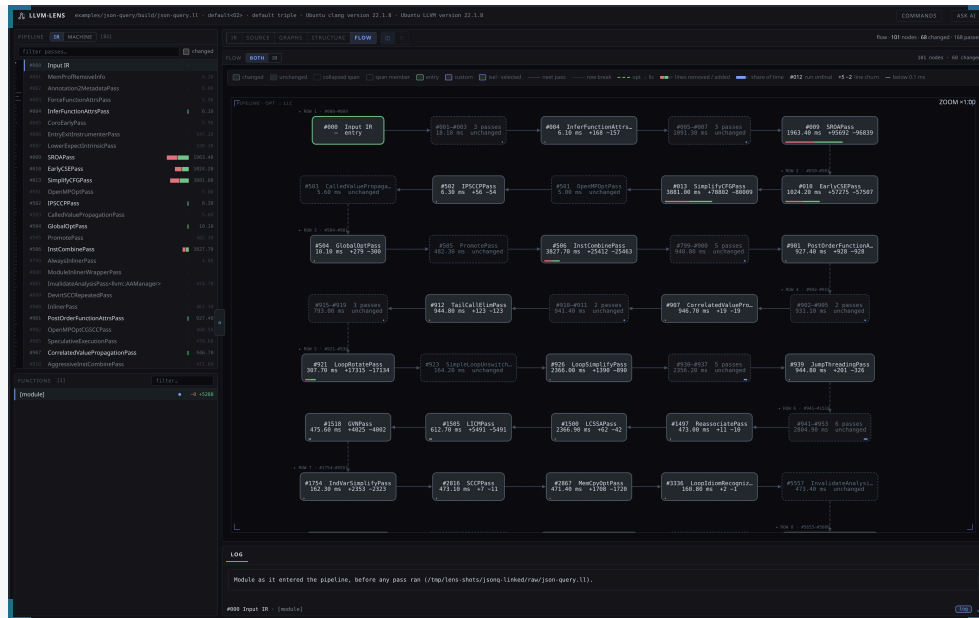


FIGURE 14 – ONE MODULE, SEVEN TRANSLATION UNITS, 101 FLOW NODES

The whole-program `default<02>` pipeline, and the same 168 cards the single-TU report has: a bigger module does not add passes, it adds runs inside them, and the churn numbers are module-scope – `+95692 -96839` summed across `SROAPass` 's 98 runs – because at module level every function's edits land in one diff. Cost is real and we would rather say it than hide it: this report took about three minutes of tool time and writes 2.3 GB of intermediate dumps. The single-TU report of the same program is 21 seconds and 313 MB, and shows every feature the bigger one does.

4.13 Provenance — the report says how it was made

A report that is going to be attached to a bug or argued over in review needs to say what produced it. The report carries the config file it read, the plugins it loaded, and the LLVM and clang versions — and a **commands** sheet prints the verbatim argv of all three tool invocations, each with a copy button. This is the whole of what LLVM-Lens asks of LLVM, and it is what the report behind Figure 2 ran:

```
$ clang-22 -S -emit-llvm -O0 -Xclang -disable-O0-optnone -g \
-o raw/lexer.ll examples/json-query/lexer.cpp

$ opt-22 -S -passes=default<02> -print-changed=quiet -print-module-scope \
-debug-pass-manager -time-passes -o raw/opt-final.ll raw/lexer.ll

$ llc-22 -print-after-all -debug-pass=Structure -time-passes \
-o raw/final.s raw/opt-final.ll
```

Every report carries these, so a reader can reproduce an analysis exactly rather than approximately. The build flag `--no-ai` is the other half of that: a report is a directory you will copy to a colleague or attach to a bug, and a credential in it would travel with it. `--no-ai` writes it with no key and no AI panel at all. Every report in this document was built that way — except the one behind Figure 15, which needed a live key to produce a real answer and was built locally, never committed, and deleted once the figure was taken.

4.14 Ask AI — the question you would otherwise look up

There is an optional **ask AI** panel, backed by your own Anthropic or OpenAI-compatible key and off until you configure one. Its whole value is that you never have to explain what you are looking at — the panel already carries the selected pass with its run count, lane, scope, churn, spill count, functions and analyses, the selected function and its diff, a bounded window of the mapped C++ under it, and the pass log. It is the same state the panes behind it are drawn from, so the question is the question you actually have:

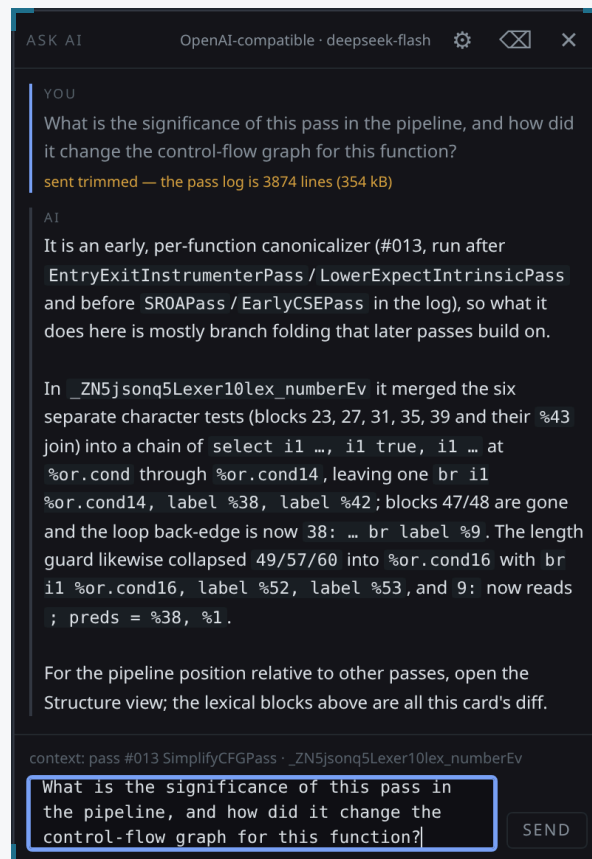


FIGURE 15 — A REAL EXCHANGE, UNEDITED

Above the answer sits the **context** line: `pass #013 SimplifyCFGPass · _ZN5jsonq5Lexer10lex_numberEv` — pass and function, the same line the panes behind are showing. Nothing in the question names either. The answer places the pass from the log (*after* `EntryExitInstrumenterPass/LowerExpectIntrinsicPass`, *before* `SR0APass/EarlyCSEPass`), then names the change: six character tests and their `%43 join` folded into one `select` chain at `%or.cond...%or.cond14`, leaving a single `br il %or.cond14`, blocks `47/48` gone, the loop back-edge renumbered to `38`, and `9`: now reading `preds = %38, %1`. Every one of those claims is checkable against the CFG pane, which is what makes the panel useful rather than merely fluent. One honest detail survives in the figure: this card's pass log is 354 kB, so the panel asked before sending it and the prompt went *trimmed*.

05 How it works

Four things are worth explaining properly: the blame walk, the two correlation layers, and the file that says how to build the report at all. The rest is parsing.

The blame walk

The dumps give us a sequence of states. Blame turns that sequence into per-line history, and the whole thing hinges on getting one judgement right: what counts as *the same line, changed* versus *a different line*.

Between two consecutive states the tool strips the common head and tail and runs an LCS diff over what is left. Each surviving line is then compared through a normalising key: SSA numbers are erased, block labels are erased, and what remains is the instruction's actual shape. So `%3 = load i64, ptr %2` becoming `%7 = load i64, ptr %5` is classified *renamed* — the same instruction after renumbering — while `%3 = load i64, ptr %2` becoming `%7 = getelementptr inbounds nuw %"class.jsonq::Lexer", ptr %0, i32 0, i32 3` is *rewritten*, and a line that was not there before is *created*.

That distinction is the difference between a history you can read and a list of every pass in the lane. Without it, every one of SROA's 28 runs and SimplifyCFG's 45 would claim whatever line it renumbered, and the chain would be noise. The chains are interned — pass names, events and history links are stored once and referenced — so the report ships the final state of the module as text rather than every intermediate state it walked through.

Instruction selection, without guessing

Correlating IR to Machine IR could have been done by matching instruction order and hoping. It isn't, because `llc` already leaves better evidence than that: each machine basic block is named after the IR block it came from, and machine operands that came from IR values carry `%ir.N` markers. The tool reads those names instead of inferring the correspondence, and reports coverage — `4/4 machine blocks traced · 6 value refs` — so a partial mapping is visibly partial.

Machine IR to assembly

Assembly is where machine instructions end up as text, and the printer does not preserve one-to-one ordering. The tool aligns the two by longest common subsequence over an opcode-compatibility predicate — `JCC_1` with `ja`, `SETCC` with `sete`, `CMOV` with `cmov` — which tolerates the assembler's spelling without pretending the two sequences are identical. What it cannot pair, it leaves unpaired and counts.

What LLVM computes, and what we compute

Worth being precise about. Every pass list, every diff, every line of IR and Machine IR, every timing and every analysis name comes from LLVM's own output. We parse it, order it, join it, and lay it out. The blame chains, the ISel and assembly correlation, the analysis graphs (dominators, dependence, loop nesting, the call graph), and the flow view's collapsing are ours — computed in Python, from the IR that `opt` and `llc` printed.

Two consequences. First, a bug in LLVM's dump is a bug in our report. Second, the report can be wrong about *interpretation* but never about *content*: the IR text in the diff is the text `opt` wrote, character for character.

Saying it once: the settings file

The flags are the interface. They are not the workflow. Nobody retypes `--load-pass-plugin ./libMBAAdd.so --custom-pass mba-add --no-ai` every time they rebuild the report they are actually working on, and most of what you want to say is not a flag at all. So LLVM-Lens also reads its settings from a file.

It looks for `llvm-lens.yml` — or `.yaml`, or `.llvm-lens.yml` — in the working directory and then in every parent, and falls back to `~/.config/llvm-lens/config.yml`; `LLVM_LENS_SETTINGS` or `--config` replaces the search entirely, and `--no-config` turns it off. Paths written in the file resolve against the file rather than your shell, so a config can be committed beside the code it analyses and behave the same from any directory. `output` is the one exception and stays relative to where you ran the command. Every key is optional, and a flag always wins over the file — the file is where you put what you already know, and the flag is for the one thing that is different today.

```
# examples/mba-add/llvm-lens.yml — the whole file
plugins:
  dir: .                                # bare names resolve here, so no loader path needed
  pass: [libMBAAdd.so]
  custom-passes: [mba-add]              # appended as function(mba-add), and badged
  ai: false                             # a report directory is meant to be shared
  ui:
    changed-only: true                  # the passes that changed nothing are noise
```

With that file in place, the entire command is `llvm-lens demo.ll`. The toolchain lives there too — which LLVM version you are on, where its binaries are, the target triple — along with the pipeline string, extra arguments for each of the three invocations, the timeout, which plugin kind you are loading, and whether source correlation is worth its extra `llc` run.

The `ui:` block is the part that saves the most, and it has no flag equivalent at all. It sets the report's own opening state: which lane and which view, which analysis graphs to draw, the split direction and ratio, which drawer tab is open, and whether the pass list starts filtered to the passes that changed something. A report you rebuild ten times a day opens ten times already pointed at the question you are asking, instead of at the default view of a pipeline you have read a hundred times.

Two details make it safe to leave lying around. An unknown key is refused with the nearest real key suggested, rather than silently ignored — the failure mode you would otherwise hit is a plausible-looking file that does nothing. And `llvm-lens --init-config` writes a fully commented starter with every key at the value it already has, so the file is discoverable without reading the source. The report carries the config it read in its own metadata, so a report that surprises you still says where its settings came from.

Why the output is plain files

The report is `index.html` plus a `data/` directory of JSON chunks, with a copy of the frontend. Each chunk is written twice — as `.json` and as a `.js` that assigns the same object to a global — so the page works when opened straight off disk with `file://`, where `fetch` is not allowed. No server, no CORS, and a report you can

zip and email. It also means the report has no runtime dependency on this project at all: it is a document.

06 Repo, team, and reproducing this

Everything in this document was produced by the code in the repository, on 18 September 2026, on a stock LLVM 22.

Running it

```
# needs python 3.10+ and llvm 22 tools on PATH
git clone https://github.com/manasghandat/LLVM-Lens.git
cd LLVM-Lens
pip install -e .
llvm-lens --sample --open                # a report, no input file needed
llvm-lens yourfile.cpp --passes 'default<02>' -o report
```

Five samples ship with the package — `licm`, `register-pressure`, `side-channel`, `switch-lowering`, `vectorize` — each written so one behaviour is legible. `examples/json-query/` is a working JSON parser in seven translation units; `examples/mba-add/` is an out-of-tree pass plugin with the config file that wires it into the report.

Before you type that second command twice, run `llvm-lens --init-config`: it writes a commented `llvm-lens.yml` listing every setting at the value it already has, and from then on the pipeline, the toolchain, the plugins and the report's opening view come from the file instead of your shell history. Section 05 explains what goes in it; it is the difference between a tool you invoke and a tool that already knows what you are working on.

GPL-3.0. Built for SEGFAULT Compiler Hackathon 2026. We used AI extensively to iterate on and develop this project. The problem statement asked for an explainable, evidence-backed view of LLVM's optimization pipeline — which passes ran, what they changed, how significant each change was, and how it affected final codegen. That is what is in the repository, and what is in the figures above.